



Two nomograms for differentiating mass-forming chronic pancreatitis from pancreatic ductal adenocarcinoma in patients with chronic pancreatitis

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Abstract

Objectives To develop and validate a CT nomogram and a radiomics nomogram to differentiate mass-forming chronic pancreatitis (MFCP) from pancreatic ductal adenocarcinoma (PDAC) in patients with chronic pancreatitis (CP).

Methods In this retrospective study, the data of 138 patients with histopathologically diagnosed MFCP or PDAC treated at our institution were retrospectively analyzed. Two radiologists analyzed the original cross-sectional CT images based on predefined criteria. Image segmentation, feature extraction, and feature reduction and selection were used to create the radiomics model. The CT and radiomics models were developed using data from a training cohort of 103 consecutive patients. The models were validated in 35 consecutive patients. Multivariable logistic regression analysis was conducted to develop a model for the differential diagnosis of MFCP and PDAC and visualized as a nomogram. The nomograms' performances were determined based on their differentiating ability and clinical utility.

Results The mean age of patients was 53.7 years, 75.4% were male. The CT nomogram showed good differentiation between the two entities in the training (area under the curve [AUC], 0.87) and validation (AUC, 0.94) cohorts. The radiomics nomogram showed good differentiation in the training (AUC, 0.91) and validation (AUC, 0.93) cohorts. Decision curve analysis showed that patients could benefit from the CT and radiomics nomograms, if the threshold probability was 0.05–0.85 and > 0.05, respectively.

Conclusions The two nomograms reasonably accurately differentiated MFCP from PDAC in patients with CP and hold potential for refining the management of pancreatic masses in CP patients.

Key Points

- A CT nomogram and a computed tomography-based radiomics nomogram reasonably accurately differentiated mass-forming chronic pancreatitis from pancreatic ductal adenocarcinoma in patients with chronic pancreatitis (CP).
- The two nomograms can monitor the cancer risk in patients with CP and hold promise to optimize the management of pancreatic masses in patients with CP.

Keywords Tomography, X-ray computed · Nomograms · Carcinoma, pancreatic ductal · Pancreatitis, chronic · Pancreatic neoplasms

Abbreviations

AIC Akaike information criterion
AUC Area under the curve

BMI Body mass index
CI Confidence interval
CT Computed tomography
DCA Decision curve analysis
DECT Dual-energy computed tomography
DWI Diffusion-weighted imaging
ICC Intraclass correlation coefficient
LASSO Least absolute shrinkage and selection operator
MFCP Mass-forming chronic pancreatitis
MRI Magnetic resonance imaging
OR Odds ratio
OS Overall survival

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PDAC	Pancreatic ductal adenocarcinoma
PV	Predictive value
ROC	Receiver operating characteristic
SE	Standard error

Introduction

Chronic pancreatitis (CP) is defined as a prolonged pancreatic inflammation that results in irreversible damage to the exocrine parenchyma and fibrosis [1]. CP is characterized morphologically by abundant fibrosis that replaces the destroyed parenchyma, leading to pancreatic atrophy [1]. However, CP may sometimes also cause enlargement of the pancreas. If this enlargement is focal, called mass-forming CP (MFCP), it can mimic pancreatic ductal adenocarcinoma (PDAC). A focal enlargement of the pancreas is seen in up to 30% of patients with CP [2].

Moreover, the risk of developing pancreatic cancer is markedly increased in patients with CP compared to those with a normal pancreas. A recent meta-analysis that included 22 studies found an increased relative risk of 13.3 folds in CP patients [3]. The clinical manifestations, risk factors, imaging characteristics, and laboratory findings overlap between PDAC and MFCP, making the distinction between these two entities often challenging. However, it is critical to differentiate MFCP from PDAC since the prognosis and treatment approaches differ substantially [4]. A delayed diagnosis may result in the progression of an early-stage PDAC that may rapidly become unresectable. On the other hand, the misdiagnosis of MFCP as PDAC may result in unnecessary surgical procedures with a high morbidity rate of 40–50% [5].

Several previous studies have proven that dual-energy computed tomography (DECT), perfusion CT, and diffusion-weighted imaging (DWI) of magnetic resonance imaging (MRI) can be utilized to distinguish MFCP from PDAC [2, 6, 7]. However, the differentiation of these two entities remains challenging, with the overall failure rate to distinguish MFCP from pancreatic cancer being as high as 25% [8]. In addition, DECT, perfusion CT, and DWI require special sequences and are difficult to be used widely. Currently, radiomics can transform the imaging data into a high-dimensional feature space and use the space to predict the tumor stage, grading, therapeutic effect, and prognosis in PDAC [9, 10]. However, few radiomics studies have documented the differential diagnosis between MFCP and PDAC [11, 12]. To the best of our knowledge, no study has documented the method of differentiating MFCP from PDAC in patients with CP.

In this study, we developed and validated a CT nomogram and a CT-based radiomics signature to differentiate MFCP from PDAC in patients with CP, and assessed the value when applied with conventional multidetector CT.

Materials and methods

Patients

The protocol for this study was approved by the Biomedical Research Ethics Committee of our institution. The need for written informed consent from the patients was waived because of the retrospective nature of the study. This study followed the Transparent Reporting of a multivariable prediction model for Individual Prognosis Or Diagnosis (TRIPOD) reporting guidelines [13].

We searched the electronic database of our institution for patients diagnosed with CP according to the Asia-Pacific consensus between February 2011 and February 2021 [14]. A total of 5433 patients were included. The exclusion criteria were as follows: (1) surgical resection or biopsy not performed; (2) histopathological diagnosis of groove pancreatitis or autoimmune pancreatitis; (3) no evaluation by contrast-enhanced multidetector CT within 1 month before surgery or biopsy; (4) pancreatic space-occupying pathological changes not visualized on multidetector CT that, therefore, could not be assessed; (5) pancreatic duct or bile duct stent; and (6) radiotherapy, chemotherapy, or chemoradiotherapy before the imaging studies. Based on these criteria, we excluded 5295 patients.

We finally enrolled 67 patients (53 patients had undergone surgery and 14 biopsies) with MFCP, including 54 men (mean age 50.31 ± 12.01 years; range 24–77 years) and 13 women (mean age 55.00 ± 12.68 years; range 35–77 years) and 71 patients (65 patients had undergone surgery and 6 biopsies) with PDAC, including 50 men (mean age 55.82 ± 11.72 years; range 36–85 years) and 21 women (mean age 56.62 ± 13.68 years; range 35–87 years). All patients in this study had a long history of CP and were receiving treatment at our institution. No pancreatic masses were found at CP diagnosis, and all masses developed in the follow-up period. The patient enrollment process for this study is shown in Fig. 1.

In the next step, we categorized the 67 patients with MFCP and 71 patients with PDAC into 103 training cases (44 MFCP cases and 59 PDAC cases), who were admitted between February 2011 and April 2018, and 35 validation cases (23 MFCP cases and 12 PDAC cases), who were admitted between May 2018 and February 2021.

Computed tomography

As a routine, multiphasic CT had been performed following a pancreas-specific protocol and using 256- and 320-slice multidetector-row CT scanners (256 slices: Brilliance-16P, Philips Healthcare; 320 slices: Aquilion ONE, Canon Medical Systems) in all patients. The CT scanning parameters were as follows: 120 kV, 150 effective mAs, beam

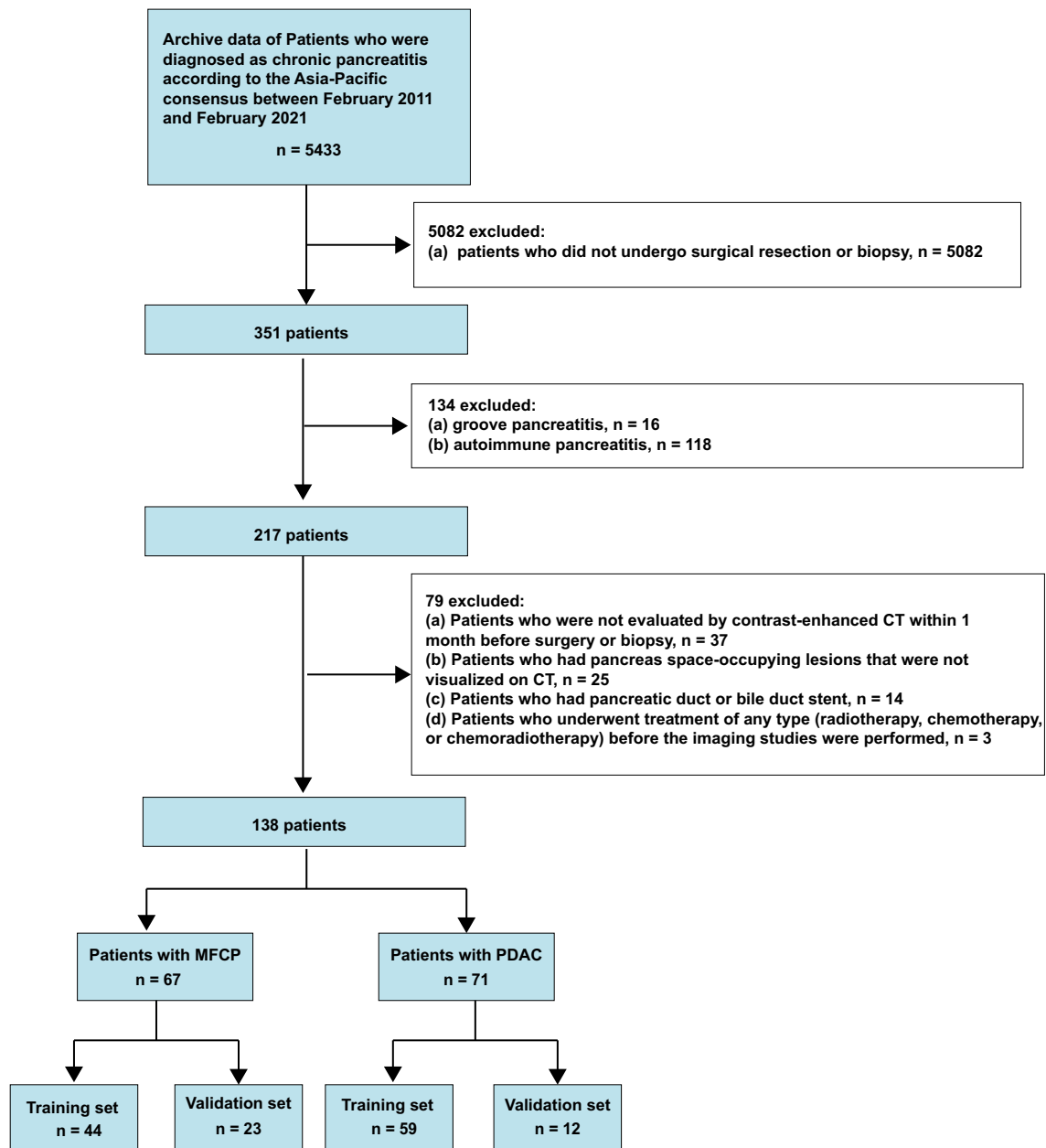


Fig. 1 Flowchart illustrating the patient selection process. PDAC: pancreatic ductal adenocarcinoma, MFCP: mass-forming chronic pancreatitis

collimation of 160×0.5 mm, a matrix of 350×350 , and a gantry rotation time of 0.5 s. A non-enhanced CT had been performed first, followed by a dynamic contrast-enhanced CT scan. The scan delay time had been determined based on the test bolus administered. The contrast agent (Ultravist 370, Bayer Schering Pharma) had been injected at a rate of 5.5 mL/s via a forearm vein, followed by 98 mL of normal saline to flush the tube. All contrast-enhanced CT scans had been performed in the arterial (20–25 s), portal venous (60–70 s), and delayed (110–130 s) phases after contrast agent injection. The slice thickness/intervals of the scan and reconstruction were 0.8/1.0 mm and 1.0/1.0 mm, respectively. The scanning range extended from the level of the diaphragm to that of the pelvis.

Radiological imaging analysis

We used the original cross-sectional CT images for the purpose of this study. All images were analyzed by two abdominal radiologists with 10 and 20 years of experience who were blinded to the clinical and pathological details. The final assessment was arrived at by consensus. All masses were evaluated for the following characteristics: (1) the location of the mass (head, neck or body, tail of the pancreas); (2) the size of the mass (the maximum cross-sectional diameter of the mass) [15]; (3) cystic degeneration (an enhancing solid portion in less than 50% area of the tumor was regarded as cystic degeneration) [16]; (4) calcification visible during unenhanced phase CT; (5) margin (well-

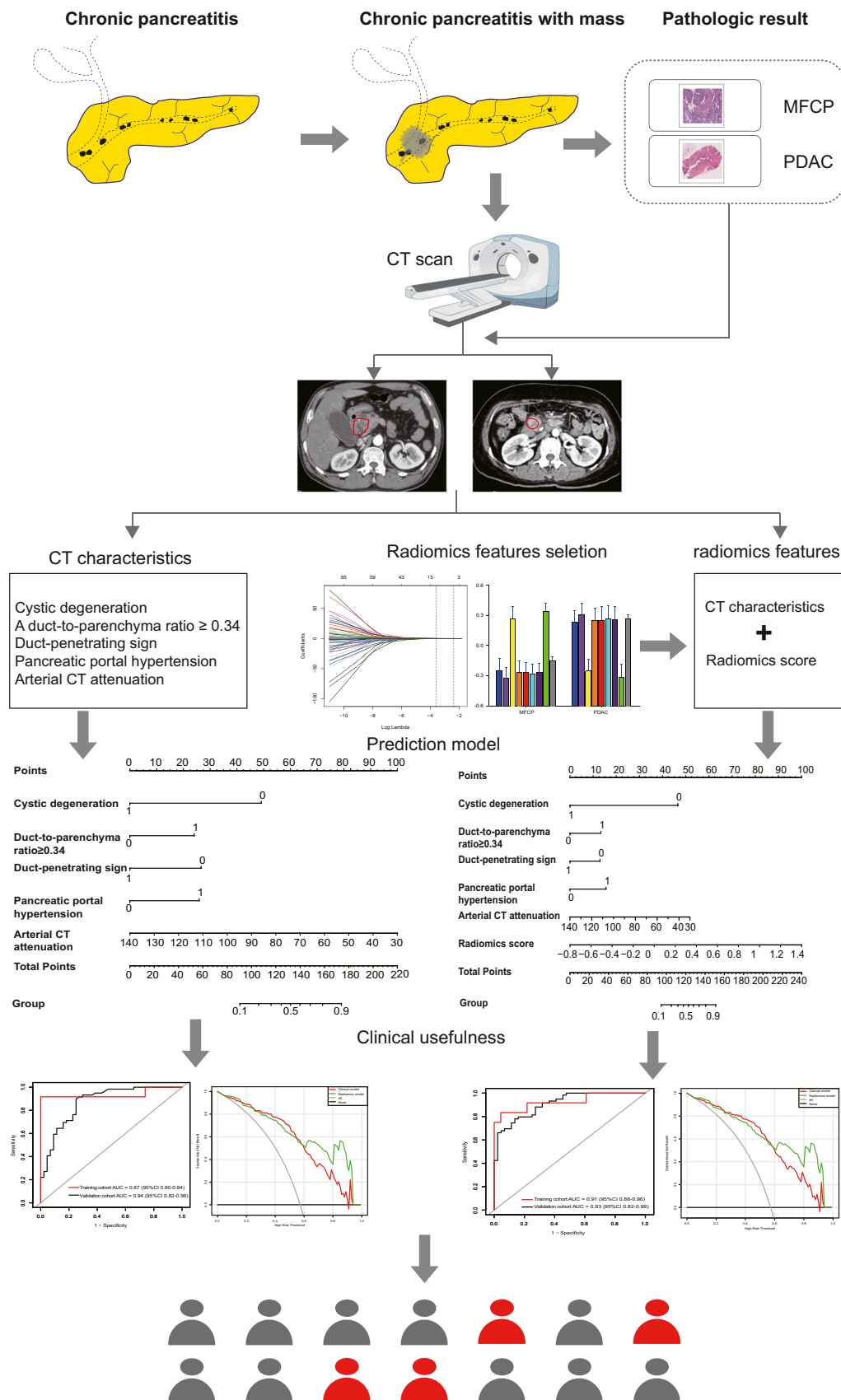


Fig. 2 The workflow of the two nomograms

delineated, implying that the tumor margin was smooth and clearly visible, or ill-delineated, implying spiculation or infiltration of more than 90° of the tumor circumference); (6) duct-to-pancreas ratio ≥ 0.34 (the pancreatic duct diameter and the pancreatic diameter were measured at the level of the splenic/portal vein confluence) [17]; (7) duct-penetrating sign (smooth narrowing of the pancreatic duct that traverses a pancreatic mass without abrupt obstruction) [18]; (8) double-duct sign; (9)

vascular invasion (the criteria were: vessel occlusion, stenosis, or more than half of the perimeter being in contact with the tumor); (10) pancreatic duct cutoff; (11) pancreatic portal hypertension (splenomegaly and varices caused by pancreatic disease); (12) peripancreatic exudation; (13) lymphadenomegaly (short-axis diameter of lymphoglandula ≥ 1 cm) [19]; and (14) CT attenuation of the mass in the non-contrast, arterial, portal venous, and delayed phases.

Table 1 Baseline characteristics of patients with MFCP and PDAC

Characteristics	Training cohort			Validation cohort		
	MFCP <i>n</i> = 44	PDAC <i>n</i> = 59	<i>p</i> value	MFCP <i>n</i> = 23	PDAC <i>n</i> = 12	<i>p</i> value
Age, years (mean $\pm$ SD)	51.07 $\pm$ 12.85	55.14 $\pm$ 10.84	0.09	51.52 $\pm$ 11.07	60.58 $\pm$ 14.67	0.05
Sex, <i>n</i> (%)			0.75			0.02
Male	34 (77.27)	44 (74.58)		20 (86.96)	6 (50.00)	
Female	10 (22.73)	15 (25.42)		3 (13.04)	6 (50.00)	
BMI, kg/m ² (mean $\pm$ SD)	20.34 $\pm$ 3.75	19.07 $\pm$ 2.15	0.92	21.37 $\pm$ 2.98	21.74 $\pm$ 3.41	0.84
Location, <i>n</i> (%)			0.12			0.88
Head and neck	24 (54.55)	41 (69.49)		14 (60.87)	7 (58.33)	
Body and tail	20 (45.45)	18 (30.51)		9 (39.13)	5 (41.67)	
Margin, <i>n</i> (%)			0.88			0.69
Ill-delineated	33 (75.00)	45 (76.27)		15 (65.22)	7 (58.33)	
Well-delineated	11 (25.00)	14 (23.73)		8 (34.78)	5 (41.67)	
Size, cm (mean $\pm$ SD)	3.51 $\pm$ 1.65	3.58 $\pm$ 1.66	0.83	3.42 $\pm$ 1.25	4.15 $\pm$ 2.15	0.21
Cystic degeneration, <i>n</i> (%)			0.01			0.45
No	28 (63.64)	50 (84.75)		16 (69.57)	10 (83.33)	
Yes	16 (36.36)	9 (15.25)		7 (30.43)	2 (16.67)	
Calcification, <i>n</i> (%)			0.11			0.11
No	25 (56.82)	24 (40.68)		16 (69.57)	5 (41.67)	
Yes	19 (43.18)	35 (59.32)		7 (30.43)	7 (58.33)	
Duct-to-pancreas ratio ≥ 0.34 , <i>n</i> (%)			0.0001			0.004
No	33 (75.00)	23 (38.98)		19 (82.61)	4 (33.33)	
Yes	11 (25.00)	36 (61.02)		4 (17.39)	8 (66.67)	
Duct-penetrating sign, <i>n</i> (%)			0.0001			0.0001
No	17 (38.64)	48 (81.36)		5 (21.74)	12 (100.00)	
Yes	27 (61.36)	11 (18.64)		18 (78.26)	0	
Double duct sign, <i>n</i> (%)			0.19			0.26
No	33 (75.00)	37 (62.71)		16 (69.57)	6 (50.00)	
Yes	11 (25.00)	22 (37.29)		7 (30.43)	6 (50.00)	
Pancreatic duct cutoff, <i>n</i> (%)			0.0001			0.0001
No	32 (72.73)	11 (18.64)		20 (86.96)	1 (8.33)	
Yes	12 (27.27)	48 (81.36)		3 (13.04)	11 (91.67)	
Pancreatic portal hypertension, <i>n</i> (%)			0.01			0.01
No	39 (88.64)	40 (67.80)		22 (95.65)	7 (58.33)	
Yes	5 (11.36)	19 (32.20)		1 (4.35)	5 (41.67)	
Peripancreatic exudation, <i>n</i> (%)			0.17			0.90
No	20 (45.45)	19 (32.20)		12 (52.17)	6 (50.00)	
Yes	24 (54.55)	40 (67.80)		11 (47.83)	6 (50.00)	
Lymphadenomegaly, <i>n</i> (%)			0.11			0.09
No	32 (72.73)	34 (57.63)		18 (78.26)	6 (50.00)	
Yes	12 (27.27)	25 (42.37)		5 (21.74)	6 (50.00)	
Nonenhanced CT attenuation, Hu (mean $\pm$ SD)	39.43 $\pm$ 9.10	37.47 $\pm$ 6.41	0.23	40.78 $\pm$ 8.82	36.67 $\pm$ 5.14	0.15
Arterial CT attenuation, Hu (mean $\pm$ SD)	71.89 $\pm$ 21.40	54.58 $\pm$ 13.67	0.0001	71.70 $\pm$ 12.82	50.08 $\pm$ 8.67	0.0001
Portal venous CT attenuation, Hu (mean $\pm$ SD)	98.64 $\pm$ 32.59	75.57 $\pm$ 21.89	0.0001	102.26 $\pm$ 19.14	65.83 $\pm$ 13.48	0.0001
Delayed CT attenuation, Hu (mean $\pm$ SD)	89.82 $\pm$ 25.08	80.64 $\pm$ 19.65	0.04	95.74 $\pm$ 17.35	71.83 $\pm$ 12.71	0.0001
Vascular invasion, <i>n</i> (%)			0.02			0.01
No	23 (52.27)	17 (28.81)		12 (52.17)	1 (8.33)	
Yes	21 (47.73)	42 (71.19)		11 (47.83)	11 (91.67)	

Note: *BMI* body mass index, *MFCP* mass-forming chronic pancreatitis, *PDAC* pancreatic ductal adenocarcinoma

Radiomics workflow

The radiomics workflow included (1) image segmentation, (2) feature extraction, and (3) feature reduction and selection. The detailed workflow is shown in Fig. 2.

Image segmentation: We used the drawing tool, which is available in the editor module of the 3D Slicer, version 4.8.1 (<https://www.slicer.org/>), to delineate the tumors in multiple slices. We extracted the volume of interest for each patient by stacking the corresponding regions of interest, delineated slice-by-slice.

Feature extraction: Radiomics feature extraction was performed using the open-source Python package Pyradiomics, version 1.2.0 (<http://www.radiomics.io/pyradiomics.html>) [20]. We used two classes of feature extraction methods: the original feature class and the filter class, which in turn included seven categories: logarithm, exponential, gradient, square, square root, lbp-2D, and wavelet. A total of 1409 2D and 3D features from masses in the portal venous phase were extracted and categorized into seven groups: (1) first-order statistics, (2) shape features, (3) gray-level co-occurrence matrix features, (4) gray-level dependence matrix features, (5) gray-level run-length matrix features, (6) gray-level size-zone matrix features, and (7) neighborhood gray-zone difference matrix features.

Feature reduction and selection: The feature selection comprised three steps: variance analysis, Spearman's correlation analysis, and the least absolute shrinkage and selection operator (LASSO) logistic regression algorithm. A retrospective power analysis was performed. The sequential method of Bonferroni's correction was applied to adjust the baseline significance level ($\alpha = 0.05$) for multiple testing bias [21, 22]. Finally, the radiomics scores (rad scores) were calculated for each patient using a linear combination of selected features that were weighted by their respective coefficients.

Intraobserver and interobserver reliability: The region of interest segmentation was performed in a blinded fashion by two

radiologists to assess interobserver reliability. Reader I extracted the features three times, with an interval of 7 days between the extractions to evaluate intraobserver reliability. Reader I completed the remaining image segmentation. Readout sessions were conducted over a period of 1 month. Reliability was calculated using the intraclass correlation coefficient (ICC) [23]. Radiomics features with both intraobserver and interobserver ICC values greater than 0.75 (indicating excellent stability) were selected for subsequent investigations.

Statistical analyses

Normal distribution and variance homogeneity tests were performed on all continuous variables. Normally distributed variables were expressed as mean $\pm$ standard deviation, while non-normally distributed variables were expressed as medians and ranges. For the purpose of the statistical analysis, we distinguished the MFCP and PDAC groups.

First, we examined potential group differences in all variables. Student's *t*-test (normal distribution), Kruskal-Wallis *H* test (skewed distribution), and the chi-square test (categorical variables) were used to determine any statistical differences between the two groups.

Second, multivariable logistic regression analysis was conducted to develop a model for the differential diagnosis of MFCP and PDAC, and a nomogram was constructed. We constructed a multivariable model using the stepwise regression method based on the Akaike information criterion (AIC) to determine the best-fitting model [24]. The differentiation performance of the established models was quantified using a receiver operating characteristic (ROC) curve. The area under the ROC curve (AUC) was calculated concurrently. ROC curves were compared using the DeLong test [25].

In this study, we developed two prediction models: a CT model and a radiomics model. The CT model was developed using CT characteristics, whereas the radiomics model was developed using CT characteristics and the radiomics score. Finally, decision curve analysis (DCA) was performed by calculating the net benefits for a range of threshold probabilities to estimate the clinical utility of the nomogram.

A two-tailed *p* value of < 0.05 was considered statistically significant. All analyses were performed using the R software (version 3.3.3, The R Foundation for Statistical Computing)

Results

Clinical and CT characteristics

Table 1 summarizes the baseline information of all patients in this study. Significant differences between the MFCP and PDAC groups were observed for duct-to-parenchyma ratio ≥ 0.34 , duct-penetrating sign, pancreatic duct cutoff, pancreatic

Table 2 The radiomics features selected by the least absolute shrinkage and selection operator logistic regression algorithm

β	Radiomics name
0.027	Original_shape_Flatness
0.180	squareroot_firstorder_Minimum
−0.013	wavelet-LHL_glcml_ClusterShade
0.060	wavelet-HLL_firstorder_Mean
0.012	wavelet-HLL_glcml_Imc2
0.087	wavelet-LLH_firstorderMean
0.010	wavelet-HLH_firstorder_Maximum
−0.287	wavelet-HHL_glcml_Imc1

Radiomics score = $0.062 + 0.027 \times \text{Original_shape_Flatness} + 0.180 \times \text{squareroot_firstorder_Minimum} - 0.013 \times \text{wavelet-LHL_glcm_ClusterShade} + 0.060 \times \text{wavelet-HLL_firstorder_Mean} + 0.012 \times \text{wavelet-HLL_glcm_Imc2} + 0.087 \times \text{wavelet-LLH_firstorderMean} + 0.010 \times \text{wavelet-HLH_firstorder_Maximum} - 0.287 \times \text{wavelet-HHL_glcm_Imc1}$

portal hypertension, arterial CT attenuation, portal venous CT attenuation, delayed CT attenuation, and vascular invasion in both the training and validation cohorts. Significant differences between the MFCP and PDAC groups were also observed for cystic degeneration in the training cohort and for sex in the validation cohort. No statistically significant differences were observed between the groups for other characteristics ($p > 0.05$).

Radiomics analysis

A total of 1409 radiomics features were extracted from the portal venous phase. The interobserver ICCs ranged from 0.77 to 0.90. The intraobserver ICCs ranged from 0.80 to 0.90. The radiomics

features that did not significantly differ between the groups or did not show significant correlations with PDAC were subsequently excluded ($p = 0.98$ – 0.10). The remaining 94 radiomics features were further reduced in the LASSO regression model. Finally, the radiomics characteristics were reduced to 8 features, and we used the LASSO formula to obtain the rad score (Table 2). The rad score was significantly lower ($p < 0.001$) in the MFCP group (median -0.16 ; range -0.70 – 0.62) than that in the PDAC group (median 0.26 ; range -0.39 – 1.28) (Fig. 3).

CT model

The multivariable logistic regression analysis included cystic degeneration, duct-to-parenchyma ratio of ≥ 0.34 , duct-

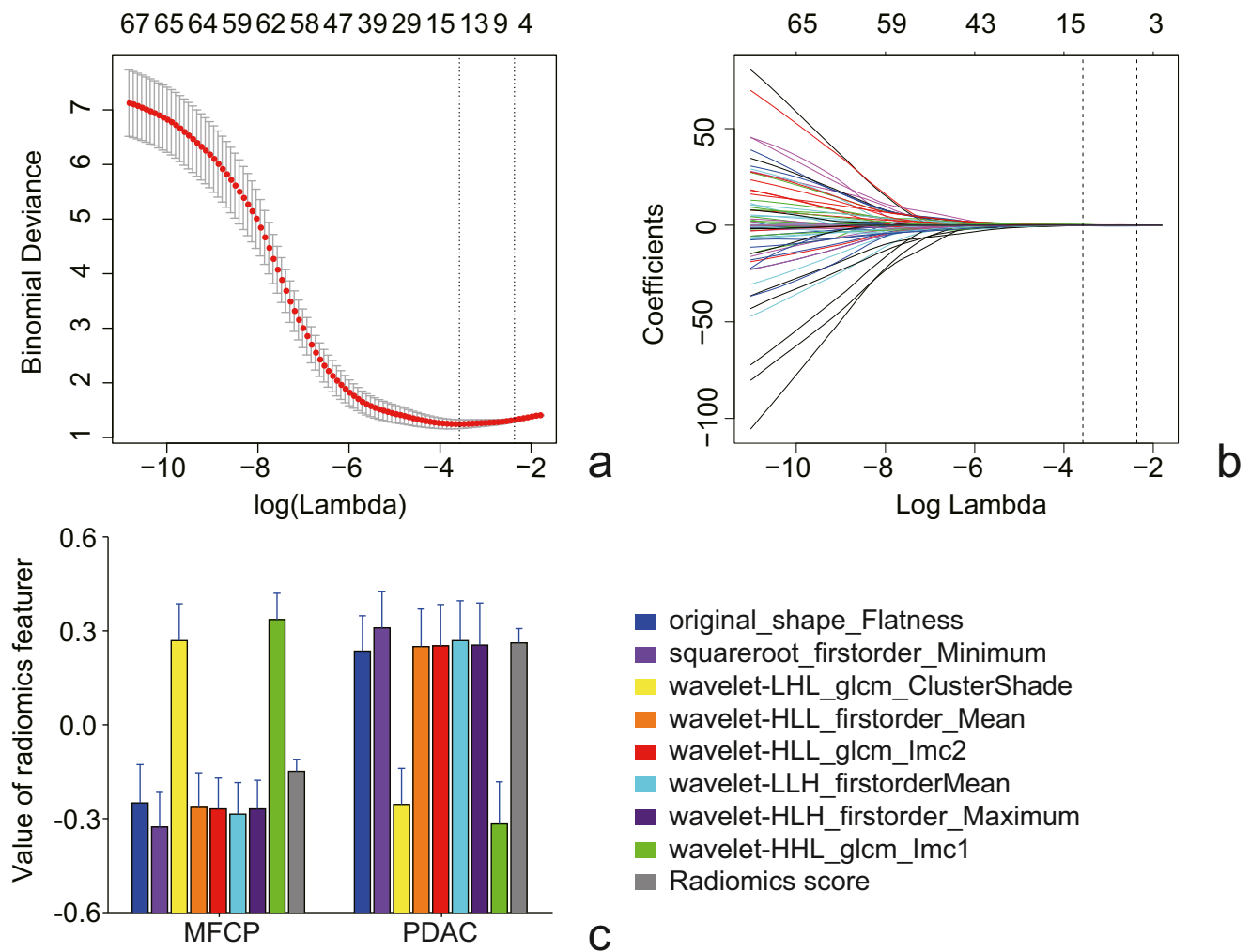


Fig. 3 Radiomics feature selection by using the parametric method, the least absolute shrinkage and selection operator (LASSO). **a** Selection of the tuning parameter (λ) in the LASSO model via 10-fold cross-validation based on minimum criteria. Binomial deviances from the LASSO regression cross-validation procedure were plotted as a function of $\log(\lambda)$. The y-axis indicates binomial deviances. The lower x-axis indicates the $\log(\lambda)$. Numbers along the upper x-axis represent the average number of predictors. Red dots indicate the average deviance values for each model with a

given λ , and vertical bars through the red dots show the upper and lower values of the deviances. The vertical black lines define the optimal values of λ , where the model provides its best fit for the data. The optimal λ value of 0.028 with $\log(\lambda) = -3.576$ was selected. **b** LASSO coefficient profiles of the 94 texture features. The dotted vertical line was plotted at the selected value using 10-fold cross-validation in **a**. The eight resulting features with nonzero coefficients are indicated in the plot. **c** The error-bar chart of the eight radiomics features and radiomics score

Table 3 The multivariable logistic regression of CT model

Variable	Coefficient	S.E.	OR (95% CI)	<i>p</i> value
(Intercept)	3.65	1.14	38.45 (4.14, 356.74)	0.001
Cystic degeneration No versus yes	−2.59	0.72	0.08 (0.02, 0.31)	0.0003
Duct-to-parenchyma ratio ≥ 0.34 No versus yes	1.26	0.60	3.54 (1.08, 11.57)	0.04
Duct-penetrating sign No versus yes	−1.40	0.57	0.25 (0.08, 0.75)	0.01
Pancreatic portal hypertension No versus yes	1.36	0.76	3.91 (0.89, 17.19)	0.07
Arterial CT attenuation	−0.05	0.02	0.95 (0.93, 0.98)	0.002

The predicted model = $3.65 - 2.59 \times (\text{cystic degeneration}) + 1.26 \times (\text{duct-to-parenchyma ratio} \geq 0.34) - 1.40 \times (\text{duct-penetrating sign}) + 1.36 \times (\text{pancreatic portal hypertension}) - 0.05 \times \text{arterial CT attenuation}$

Note: *S.E.* standard error, *OR* odds ratio, *CI* confidence interval

penetrating sign, pancreatic duct cutoff, pancreatic portal hypertension, arterial CT attenuation, portal venous CT attenuation, delayed CT attenuation, and vascular invasion. Consequently, cystic degeneration ($p < 0.001$), duct-to-parenchyma ratio ≥ 0.34 ($p = 0.04$), duct-penetrating sign ($p = 0.01$), pancreatic portal hypertension ($p = 0.07$), and arterial CT attenuation ($p = 0.002$) were selected for the CT model (Table 3 and Fig. 4a). The performance of the prediction model is shown in Fig. 4b.

Radiomics model

The multivariable logistic regression analysis included cystic degeneration, duct-to-parenchyma ratio of ≥ 0.34 , duct-penetrating sign, pancreatic duct cutoff, pancreatic portal hypertension, arterial CT attenuation, portal venous CT attenuation, delayed CT attenuation, vascular invasion, and radiomics score. Consequently, cystic degeneration ($p < 0.001$), duct-to-parenchyma ratio ≥ 0.34 ($p = 0.13$), duct-penetrating sign ($p = 0.13$), pancreatic portal hypertension ($p = 0.15$), arterial CT attenuation ($p = 0.03$), and radiomics score ($p = 0.001$) were selected for the radiomics model (Table 4 and Fig. 4c). The performance of the prediction model is shown in Fig. 4d and Table 5. The cases diagnosed using the two nomograms are shown in Fig. 5. There was not a significant difference between the radiomics model and the CT model ($p = 0.09$).

Clinical utility

The DCA showed that if the threshold probability was 0.05–0.85, using our CT model to distinguish MFCP from PDAC in patients with CP was more beneficial than the “treat-all-patients as MFCP” scheme or the “treat-all-patients as PDAC” scheme. The DCA also showed that if the threshold

probability was > 0.05 , using our radiomics model to distinguish MFCP from PDAC in patients with CP was more beneficial than the “treat-all-patients as MFCP” scheme or the “treat-all-patients as PDAC scheme” (Fig. 6).

Discussion

In this study, we developed and validated a CT nomogram and a radiomics nomogram to differentiate MFCP from PDAC in patients with CP. To our knowledge, this is the first study to investigate the use of the conventional CT characteristics and the CT radiomics features to discriminate MFCP from PDAC in patients with CP. We found that the two nomograms showed good performance and may be useful in prognostic prediction and decision-making for the need for surgical resection.

Previous studies have summarized the radiologic characteristics indicating MFCP rather than malignancy, which are parenchymal calcifications, pseudocysts, duct-penetrating sign, and pancreatic duct-to-parenchyma ratio of less than 0.34 [17, 18, 26, 27]. Our study confirmed these results except for the parenchymal calcifications, which may be explained by the long history of CP in all our patients, a disease characterized by parenchymal calcification. In addition, we also found that the CT attenuations in the arterial, portal venous, and delayed phases were statistically significantly higher in MFCP than PDAC. This finding is supported by several studies that indicated that the density of microvascular structures is lower in PDAC than MFCP and that intra-tumoral blood vessel infiltration and fibrosis are more pronounced in PDAC than in MFCP [6, 28].

Several previous studies have proven that dual-energy CT, perfusion CT, and diffusion-weighted MRI can be utilized to

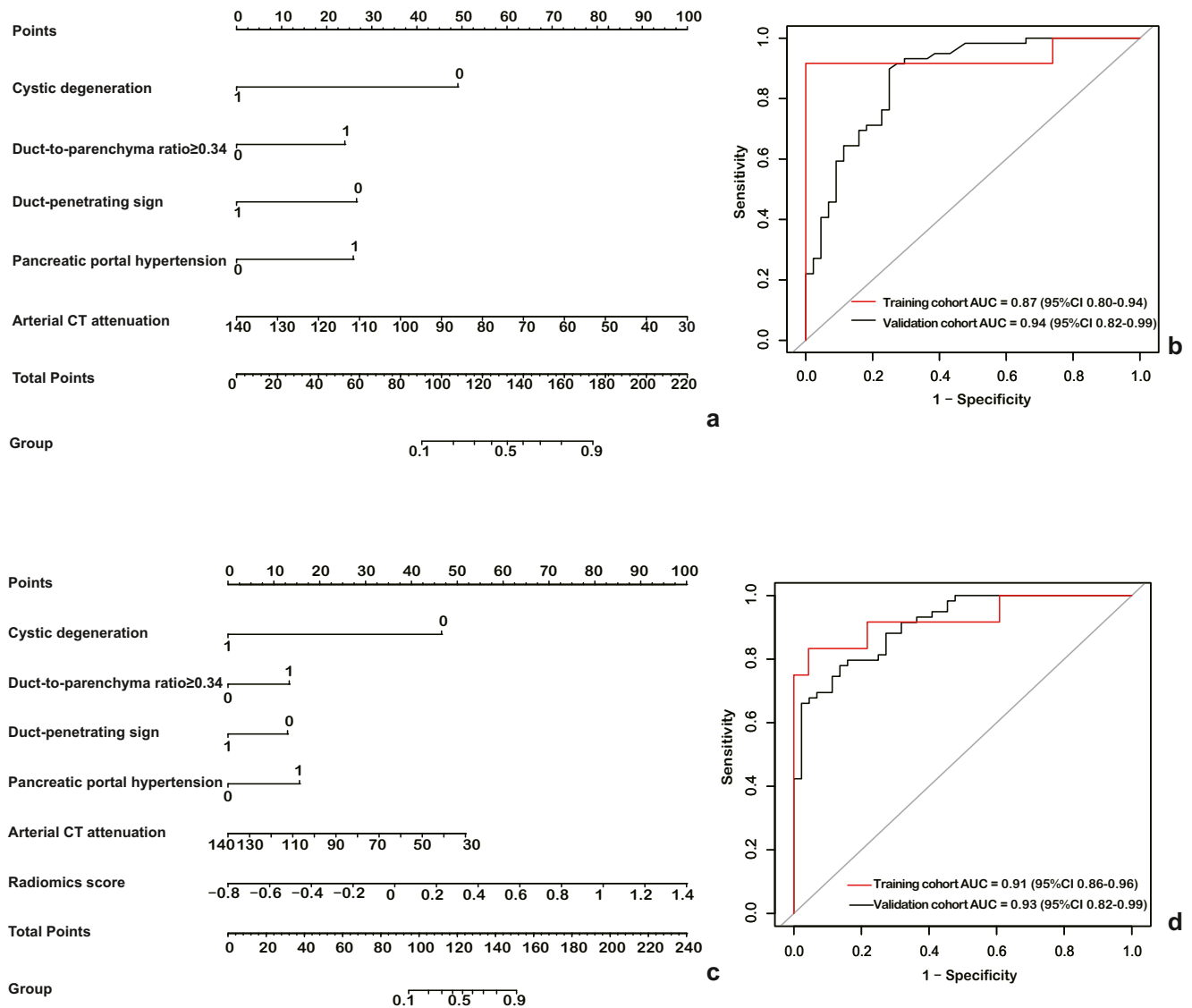


Fig. 4 **a** A CT nomogram in the training cohort. **b** Receiver-operating characteristic (ROC) curves of the CT nomogram in the training and validation cohorts. **c** A radiomics nomogram in the training cohort. **d**

Receiver-operating characteristic (ROC) curves of the radiomics nomogram in the training and validation cohorts

Table 4 The multivariable logistic regression of radiomics model

Variable	Coefficient	S.E.	OR (95% CI)	<i>p</i> value
(Intercept)	3.14	1.25	23.15 (1.98, 270.78)	0.01
Cystic degeneration	− 3.62	0.93	0.03 (0.004, 0.17)	0.0001
No versus yes				
Duct-to-parenchyma ratio ≥ 0.34	1.04	0.67	2.83 (0.76, 10.57)	0.13
No versus yes				
Duct-penetrating sign	− 1.01	0.66	0.36 (0.10, 1.33)	0.13
No versus yes				
Pancreatic portal hypertension	1.21	0.85	3.37 (0.63, 17.85)	0.15
No versus yes				
Arterial CT attenuation	− 0.04	0.02	0.96 (0.93, 1.00)	0.03
Radiomics score	3.54	1.08	34.39 (4.13, 286.57)	0.001

The predicted model = $3.14 - 3.62 \times (\text{cystic degeneration}) + 1.04 \times (\text{duct-to-parenchyma ratio} \geq 0.34) - 1.01 \times (\text{duct-penetrating sign}) + 1.21 \times (\text{Pancreatic portal hypertension}) - 0.04 \times \text{arterial CT attenuation} + 3.54 \text{ radiomics score}$

Note: *S.E.* standard error, *OR* odds ratio, *CI* confidence interval

Table 5 The performance of the prediction model

Performance	CT model		Radiomics model	
	Training cohort	Validation cohort	Training cohort	Validation cohort
AUC (95%CI)	0.87 (0.80 – 0.94)	0.94 (0.82–0.99)	0.91 (0.86 – 0.96)	0.93 (0.82 – 0.99)
Sensitivity (%)	89.83	91.67	77.97	83.33
Specificity (%)	75.00	100	86.36	95.65
Accuracy (%)	83.50	97.14	81.55	91.43
PPV (%)	82.81	100	88.46	90.91
NPV (%)	84.62	95.83	74.51	91.67

Note. *AUC* area under the curve, *CI* confidence interval, *PPV* positive predictive value, *NPV* negative predictive value

distinguish MFCP from PDAC, and the AUC values of the newer technologies ranged from 0.76 to 0.83 [6, 7, 28, 29]. However, the use of DECT, perfusion CT, and DWI is limited in some institutions with the overall failure rate to distinguish MFCP from PDAC being as high as 25% [8]. The value of radiomics analysis in differentiating MFCP from PDAC has been demonstrated previously. Deng et al [11] included 51 patients with PDAC and 13 patients with MFCP, and developed four radiomics models based on T1-weighted imaging, T2-weighted imaging, and the artery and portal phases of

dynamic contrast-enhanced MRI. All models achieved good performance (the AUC values of the radiomics models ranged from 0.882 to 0.997). The study by Ren et al [12] included 79 patients with PDAC and 30 patients with MFCP, and showed that the AUC of CT imaging feature-based, texture feature-based in arterial and portal phases, and the combined models were 0.84, 0.96, 0.93, and 0.98, respectively. The major limitation of the above radiomics studies was the small sample size, especially for the patients with MFCP. Besides, all patients with masses in these studies

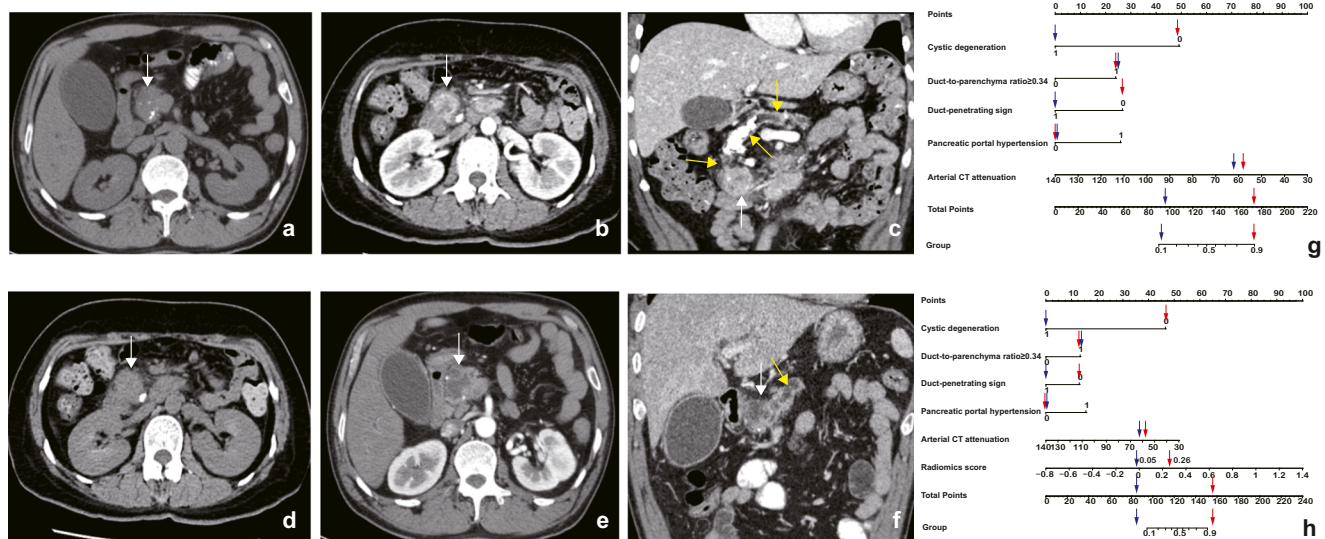


Fig. 5 The two nomograms accurately differentiate mass-forming chronic pancreatitis (MFCP) from pancreatic ductal adenocarcinoma (PDAC) in patients with chronic pancreatitis. **a–c** Case 1: a 45-year-old woman with pathologically confirmed PDAC, who was misdiagnosed with MFCP by a radiologist. **a** Axial non-enhanced computed tomography (CT) image shows an isoattenuation mass with calcification located at the pancreatic head (white arrow). **b** Axial contrast-enhanced arterial phase image demonstrates a mildly heterogeneous, hyper-attenuation mass (58 Hu) without cystic degeneration (white arrow). **c** Coronal contrast-enhanced portal phase image demonstrates the main pancreatic duct cutoff with a duct-to-parenchyma ratio ≥ 0.34 (yellow arrow), and no pancreatic portal hypertension. **d–f** Case 2: a 48-year-old man with

pathologically confirmed MFCP, who was misdiagnosed with PDAC by a radiologist. **d** Axial non-enhanced CT image shows an isoattenuation mass with calcification located at the pancreatic head (white arrow). **e** Axial contrast-enhanced arterial phase image demonstrates a heterogeneous, hypoattenuation mass (62 Hu) with cystic degeneration (white arrow). **f** Coronal contrast-enhanced portal phase image demonstrates duct-penetrating sign (yellow arrow) with a duct-to-parenchyma ratio ≥ 0.34 , and no pancreatic portal hypertension. **g** According to the CT model, the prediction probability of PDAC is 89.00% (red arrow) and 11.00% (blue arrow) in cases 1 and 2, respectively. **h** According to the radiomics model, the prediction probability of PDAC is greater than 90.00% (red arrow) and less than 10.00% (blue arrow) in cases 1 and 2, respectively

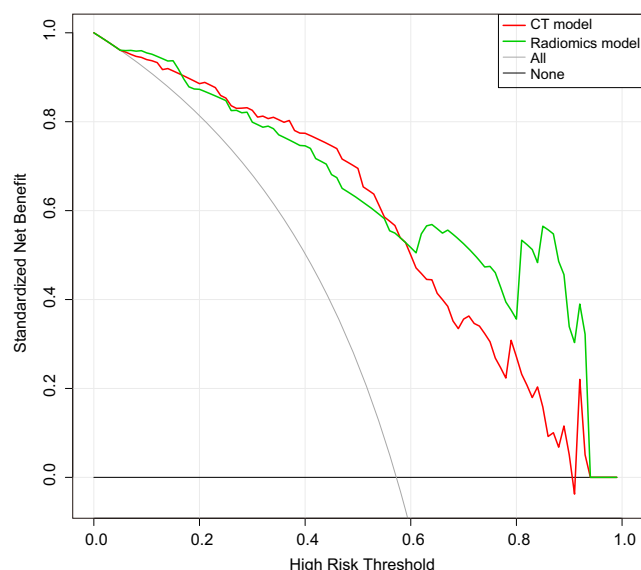


Fig. 6 Decision curve analysis for the CT and radiomics models. The y-axis represents the net benefit. The red line represents the CT model. The green line represents the radiomics model. The black line represents the hypothesis that all patients had mass-forming chronic pancreatitis (MFCP). The gray line represents the hypothesis that all patients had pancreatic ductal adenocarcinoma (PDAC). The x-axis represents the threshold probability, which is the point at which the expected benefit of treatment is equal to the expected benefit of avoiding treatment. The decision curves in the validation cohort indicated that if the threshold probability is 0.05–0.85 and > 0.05 in the CT and radiomics models, respectively, using the nomogram to predict MFCP and PDAC is more beneficial than the “treat-all-patients as MFCP” scheme or the “treat-all-patients as PDAC” scheme

did not have a history of long-standing CP. Monitoring the cancer risk in patients with CP using the above-mentioned models can be difficult. The strengths of our study are that all patients had a long history of CP. Moreover, this study used a relatively large data set including 71 patients with PDAC and 69 patients with MFCP. Furthermore, our study showed that the differentiation abilities of the two nomograms were not significantly different.

Our study has several limitations. First, one of the main drawbacks of radiomics research is that the poor interpretation of radiomics features hinders the clinical promotion of radiomics. Although many studies have attempted to investigate potential correlations considering the grayscale intensity and matrix uniformity, it remains inherently difficult to directly link the radiomics features with the clinical status. We also faced a similar problem. Second, this was a single-center retrospective study with a relatively small sample size, which may result in a selection bias. However, as far as we know, it was a large sample for a single center. In the future, a multicenter study with more patients should investigate the potential value of a CT-based radiomics model in differentiating MFCP from PDAC in patients with CP. Third, although many patients with masses not visible on CT were excluded, the

margins of some masses were difficult to define, which affects the reproducibility of our study to some extent. Consequently, our findings do not apply to masses not visible on CT but visible on MRI and endoscopic ultrasonography.

In conclusion, we developed and validated a CT model and a CT-based radiomics model. Both were shown to be reasonably accurate in their differentiation of MFCP from PDAC in patients with CP and hold substantial potential for refining the management of pancreatic masses in these patients.

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Declarations

Guarantor The scientific guarantor of this publication is Yun Bian.

Conflict of interest The authors of this manuscript declare no relationships with any companies, whose products or services may be related to the subject matter of the article.

Statistics and biometry A professor of Biostatistics (Dr Pin Wu, PhD) was consulted for specialist advice.

Informed consent Written informed consent was waived by the Institutional Review Board.

Ethical approval Institutional Review Board approval was obtained by the Changhai Hospital.

Methodology

- retrospective
- diagnostic or prognostic study
- performed at one institution

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